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Methods for compensating for optical surface nonuniformity

Abstract

Systems and methods for compensating for nonuniform surface topography features in accordance with various embodiments of the invention are illustrated. One embodiment includes a method for manufacturing waveguide cells, the method including providing a waveguide including first and second substrates and a layer of optical recording material, and applying a surface forming process to at least one external surface of the first and second substrates. In another embodiment, applying the surface forming process includes applying a forming material coating to the at least one external surface, providing a forming element having a forming surface, bringing the forming element in physical contact with the forming material coating, curing the forming material coating while it is in contact with the forming element, and releasing the forming material coating from the forming element.

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6151142	12/1999	Phillips et al.	N/A	N/A
6154190	12/1999	Yang et al.	N/A	N/A
6156243	12/1999	Kosuga et al.	N/A	N/A
6167169	12/1999	Brinkman et al.	N/A	N/A
6169594	12/2000	Aye et al.	N/A	N/A
6169613	12/2000	Amitai et al.	N/A	N/A
6169636	12/2000	Kreitzer et al.	N/A	N/A
6172792	12/2000	Jepsen et al.	N/A	N/A
6176837	12/2000	Foxlin	N/A	N/A
6185015	12/2000	Reinhorn et al.	N/A	N/A
6185016	12/2000	Popovich	N/A	N/A
6188462	12/2000	Lavrentovich et al.	N/A	N/A
6191887	12/2000	Michaloski et al.	N/A	N/A
6195206	12/2000	Yona et al.	N/A	N/A
6195209	12/2000	Kreitzer et al.	N/A	N/A
6204835	12/2000	Yang et al.	N/A	N/A
6211976	12/2000	Popovich et al.	N/A	N/A
6215579	12/2000	Bloom et al.	N/A	N/A
6218316	12/2000	Marsh	N/A	N/A
6222297	12/2000	Perdue	N/A	N/A
6222675	12/2000	Mall et al.	N/A	N/A
6222971	12/2000	Veligdan et al.	N/A	N/A
6249386	12/2000	Yona et al.	N/A	N/A
6259423	12/2000	Tokito et al.	N/A	N/A
6259559	12/2000	Kobayashi et al.	N/A	N/A
6266166	12/2000	Katsumata et al.	N/A	N/A
6268839	12/2000	Yang et al.	N/A	N/A
6269203	12/2000	Davies et al.	N/A	N/A
6275031	12/2000	Simmonds et al.	N/A	N/A
6278429	12/2000	Ruth et al.	N/A	N/A
6285813	12/2000	Schultz et al.	N/A	N/A
6297860	12/2000	Moskovich et al.	N/A	N/A
6301056	12/2000	Kreitzer et al.	N/A	N/A
6301057	12/2000	Kreitzer et al.	N/A	N/A
6317083	12/2000	Johnson et al.	N/A	N/A
6317227	12/2000	Mizutani et al.	N/A	N/A
6317228	12/2000	Popovich et al.	N/A	N/A
6317528	12/2000	Gadkaree et al.	N/A	N/A
6320563	12/2000	Yang et al.	N/A	N/A
6321069	12/2000	Piirainen	N/A	N/A
6323970	12/2000	Popovich	N/A	N/A
6323989	12/2000	Jacobson et al.	N/A	N/A
6324014	12/2000	Moskovich et al.	N/A	N/A
6327089	12/2000	Hosaki et al.	N/A	N/A
6330109	12/2000	Ishii et al.	N/A	N/A
6333819	12/2000	Svedenkrans	N/A	N/A
6335224	12/2001	Peterson et al.	N/A	N/A

6339486	12/2001	Popovich	N/A	N/A
6340540	12/2001	Ueda et al.	N/A	N/A
6351273	12/2001	Lemelson et al.	N/A	N/A
6351333	12/2001	Araki et al.	N/A	N/A
6356172	12/2001	Koivisto et al.	N/A	N/A
6356674	12/2001	Davis et al.	N/A	N/A
6359730	12/2001	Tervonen	N/A	N/A
6359737	12/2001	Stringfellow	N/A	N/A
6366281	12/2001	Lipton et al.	N/A	N/A
6366369	12/2001	Ichikawa et al.	N/A	N/A
6366378	12/2001	Tervonen et al.	N/A	N/A
6377238	12/2001	McPheters	N/A	N/A
6377321	12/2001	Khan et al.	N/A	N/A
6388797	12/2001	Lipton et al.	N/A	N/A
6392812	12/2001	Howard	N/A	N/A
6407724	12/2001	Waldern et al.	N/A	N/A
6409687	12/2001	Foxlin	N/A	N/A
6411444	12/2001	Moskovich et al.	N/A	N/A
6414760	12/2001	Lopez et al.	N/A	N/A
6417971	12/2001	Moskovich et al.	N/A	N/A
6437563	12/2001	Simmonds et al.	N/A	N/A
6445512	12/2001	Moskovich et al.	N/A	N/A
6449095	12/2001	Ohtaki et al.	N/A	N/A
6456584	12/2001	Nagata et al.	N/A	N/A
6470132	12/2001	Nousiainen et al.	N/A	N/A
6473209	12/2001	Popovich	N/A	N/A
6476974	12/2001	Kreitzer et al.	N/A	N/A
6483303	12/2001	Simmonds et al.	N/A	N/A
6486997	12/2001	Bruzzzone et al.	N/A	N/A
6504518	12/2002	Kuwayama et al.	N/A	N/A
6504629	12/2002	Popovich et al.	N/A	N/A
6509937	12/2002	Moskovich et al.	N/A	N/A
6510263	12/2002	Maisenhoelder et al.	N/A	N/A
6518747	12/2002	Sager et al.	N/A	N/A
6519088	12/2002	Lipton	N/A	N/A
6522794	12/2002	Bischel et al.	N/A	N/A
6522795	12/2002	Jordan et al.	N/A	N/A
6524771	12/2002	Maeda et al.	N/A	N/A
6529336	12/2002	Kreitzer et al.	N/A	N/A
6534977	12/2002	Duncan et al.	N/A	N/A
6538775	12/2002	Bowley et al.	N/A	N/A
6545778	12/2002	Ono et al.	N/A	N/A
6545808	12/2002	Ehbets et al.	N/A	N/A
6550949	12/2002	Bauer et al.	N/A	N/A
6552789	12/2002	Modro	N/A	N/A
6557413	12/2002	Nieminen et al.	N/A	N/A
6559813	12/2002	DeLuca et al.	N/A	N/A
6560019	12/2002	Nakai	N/A	N/A
6563648	12/2002	Gleckman et al.	N/A	N/A

6563650	12/2002	Moskovich et al.	N/A	N/A
6567014	12/2002	Hansen et al.	N/A	N/A
6567573	12/2002	Domash et al.	N/A	N/A
6577411	12/2002	David et al.	N/A	N/A
6577429	12/2002	Kurtz et al.	N/A	N/A
6580529	12/2002	Amitai et al.	N/A	N/A
6583838	12/2002	Hoke et al.	N/A	N/A
6583873	12/2002	Goncharov et al.	N/A	N/A
6587619	12/2002	Kinoshita	N/A	N/A
6594090	12/2002	Kruschwitz et al.	N/A	N/A
6596193	12/2002	Coates et al.	N/A	N/A
6597176	12/2002	Simmonds et al.	N/A	N/A
6597475	12/2002	Shirakura et al.	N/A	N/A
6598987	12/2002	Parikka	N/A	N/A
6600590	12/2002	Roddy et al.	N/A	N/A
6608720	12/2002	Freeman	N/A	N/A
6611253	12/2002	Cohen	N/A	N/A
6618104	12/2002	Date et al.	N/A	N/A
6624943	12/2002	Nakai et al.	N/A	N/A
6625381	12/2002	Roddy et al.	N/A	N/A
6646772	12/2002	Popovich et al.	N/A	N/A
6646810	12/2002	Harter et al.	N/A	N/A
6661578	12/2002	Hedrick	N/A	N/A
6667134	12/2002	Sutherland et al.	N/A	N/A
6674578	12/2003	Sugiyama et al.	N/A	N/A
6677086	12/2003	Sutehrland et al.	N/A	N/A
6678093	12/2003	Scobey et al.	N/A	N/A
6680720	12/2003	Lee et al.	N/A	N/A
6686815	12/2003	Mirshekarl-Syahkal et al.	N/A	N/A
6690516	12/2003	Aritake et al.	N/A	N/A
6692666	12/2003	Sutherland et al.	N/A	N/A
6699407	12/2003	Sutehrland et al.	N/A	N/A
6706086	12/2003	Emig et al.	N/A	N/A
6706451	12/2003	Sutherland et al.	N/A	N/A
6714329	12/2003	Sekine et al.	N/A	N/A
6721096	12/2003	Bruzzone et al.	N/A	N/A
6730442	12/2003	Sutherland et al.	N/A	N/A
6731434	12/2003	Hua et al.	N/A	N/A
6738105	12/2003	Hannah et al.	N/A	N/A
6741189	12/2003	Gibbons et al.	N/A	N/A
6744478	12/2003	Asakura et al.	N/A	N/A
6747781	12/2003	Trisnadi et al.	N/A	N/A
6748342	12/2003	Dickhaus	N/A	N/A
6750941	12/2003	Satoh et al.	N/A	N/A
6750995	12/2003	Dickson	N/A	N/A
6750996	12/2003	Jagt et al.	N/A	N/A
6757105	12/2003	Niv et al.	N/A	N/A
6771403	12/2003	Endo et al.	N/A	N/A
6776339	12/2003	Piikivi	N/A	N/A

6781701	12/2003	Sweetser et al.	N/A	N/A
6791629	12/2003	Moskovich et al.	N/A	N/A
6791739	12/2003	Ramanujan et al.	N/A	N/A
6804066	12/2003	Ha et al.	N/A	N/A
6805490	12/2003	Levola	N/A	N/A
6821457	12/2003	Natarajan et al.	N/A	N/A
6822713	12/2003	Yaroshchuk et al.	N/A	N/A
6825987	12/2003	Repetto et al.	N/A	N/A
6829095	12/2003	Amitai	N/A	N/A
6830789	12/2003	Doane et al.	N/A	N/A
6833955	12/2003	Niv	N/A	N/A
6836369	12/2003	Fujikawa et al.	N/A	N/A
6842563	12/2004	Zhang et al.	N/A	N/A
6844212	12/2004	Bond et al.	N/A	N/A
6844980	12/2004	He et al.	N/A	N/A
6844989	12/2004	Jo et al.	N/A	N/A
6847274	12/2004	Salmela et al.	N/A	N/A
6847488	12/2004	Travis	N/A	N/A
6850210	12/2004	Lipton et al.	N/A	N/A
6853491	12/2004	Ruhle et al.	N/A	N/A
6853493	12/2004	Kreitzer et al.	N/A	N/A
6861107	12/2004	Klasen-Memmer et al.	N/A	N/A
6864861	12/2004	Schehrer et al.	N/A	N/A
6864927	12/2004	Cathey	N/A	N/A
6864931	12/2004	Kumar et al.	N/A	N/A
6867888	12/2004	Sutherland et al.	N/A	N/A
6873443	12/2004	Joubert et al.	N/A	N/A
6876791	12/2004	Murashima et al.	N/A	N/A
6878494	12/2004	Sutehrland et al.	N/A	N/A
6885483	12/2004	Takada	N/A	N/A
6903872	12/2004	Schrader	N/A	N/A
6909345	12/2004	Salmela et al.	N/A	N/A
6917375	12/2004	Akada et al.	N/A	N/A
6919003	12/2004	Ikeda et al.	N/A	N/A
6922267	12/2004	Endo et al.	N/A	N/A
6926429	12/2004	Barlow et al.	N/A	N/A
6927570	12/2004	Simmonds et al.	N/A	N/A
6927694	12/2004	Smith et al.	N/A	N/A
6940361	12/2004	Jokio et al.	N/A	N/A
6943788	12/2004	Tomono	N/A	N/A
6950173	12/2004	Sutherland et al.	N/A	N/A
6950227	12/2004	Schrader	N/A	N/A
6951393	12/2004	Koide	N/A	N/A
6952312	12/2004	Weber et al.	N/A	N/A
6952435	12/2004	Lai et al.	N/A	N/A
6958662	12/2004	Salmela et al.	N/A	N/A
6958868	12/2004	Pender	N/A	N/A
6963454	12/2004	Martins et al.	N/A	N/A
6972788	12/2004	Robertson et al.	N/A	N/A

6975345	12/2004	Lipton et al.	N/A	N/A
6980365	12/2004	Moskovich	N/A	N/A
6985296	12/2005	Lipton et al.	N/A	N/A
6987908	12/2005	Bond et al.	N/A	N/A
6999239	12/2005	Martins et al.	N/A	N/A
7002618	12/2005	Lipton et al.	N/A	N/A
7002753	12/2005	Moskovich et al.	N/A	N/A
7003075	12/2005	Miyake et al.	N/A	N/A
7003187	12/2005	Frick et al.	N/A	N/A
7006732	12/2005	Gunn et al.	N/A	N/A
7009773	12/2005	Chaoulov et al.	N/A	N/A
7018563	12/2005	Sutherland et al.	N/A	N/A
7018686	12/2005	Sutehrland et al.	N/A	N/A
7018744	12/2005	Otaki et al.	N/A	N/A
7019793	12/2005	Moskovich et al.	N/A	N/A
7021777	12/2005	Amitai	N/A	N/A
7026892	12/2005	Kajiya	N/A	N/A
7027671	12/2005	Huck et al.	N/A	N/A
7034748	12/2005	Kajiya	N/A	N/A
7046439	12/2005	Kaminsky et al.	N/A	N/A
7050674	12/2005	Lee et al.	N/A	N/A
7053735	12/2005	Salmela et al.	N/A	N/A
7053991	12/2005	Sandusky	N/A	N/A
7054045	12/2005	McPheters et al.	N/A	N/A
7058434	12/2005	Wang et al.	N/A	N/A
7068405	12/2005	Sutherland et al.	N/A	N/A
7068898	12/2005	Buretea et al.	N/A	N/A
7072020	12/2005	Sutherland et al.	N/A	N/A
7075273	12/2005	O'Gorman et al.	N/A	N/A
7077984	12/2005	Natarajan et al.	N/A	N/A
7081215	12/2005	Natarajan et al.	N/A	N/A
7088457	12/2005	Zou et al.	N/A	N/A
7088515	12/2005	Lipton	N/A	N/A
7095562	12/2005	Peng et al.	N/A	N/A
7099080	12/2005	Lipton et al.	N/A	N/A
7101048	12/2005	Travis	N/A	N/A
7108383	12/2005	Mitchell et al.	N/A	N/A
7110184	12/2005	Yona et al.	N/A	N/A
7119965	12/2005	Rolland et al.	N/A	N/A
7123418	12/2005	Weber et al.	N/A	N/A
7123421	12/2005	Moskovich et al.	N/A	N/A
7126418	12/2005	Hunton et al.	N/A	N/A
7126583	12/2005	Breed	N/A	N/A
7132200	12/2005	Ueda et al.	N/A	N/A
7133084	12/2005	Moskovich et al.	N/A	N/A
7139109	12/2005	Mukawa	N/A	N/A
RE39424	12/2005	Moskovich	N/A	N/A
7145729	12/2005	Kreitzer et al.	N/A	N/A
7149385	12/2005	Parikka et al.	N/A	N/A
7151246	12/2005	Fein et al.	N/A	N/A

7158095	12/2006	Jenson et al.	N/A	N/A
7167286	12/2006	Anderson et al.	N/A	N/A
7167616	12/2006	Ling et al.	N/A	N/A
7175780	12/2006	Sutherland et al.	N/A	N/A
7181105	12/2006	Teramura et al.	N/A	N/A
7181108	12/2006	Levola	N/A	N/A
7184002	12/2006	Lipton et al.	N/A	N/A
7184615	12/2006	Levola	N/A	N/A
7186567	12/2006	Sutherland et al.	N/A	N/A
7190849	12/2006	Katase	N/A	N/A
7198737	12/2006	Natarajan et al.	N/A	N/A
7199934	12/2006	Yamasaki	N/A	N/A
7205960	12/2006	David	N/A	N/A
7205964	12/2006	Yokoyama et al.	N/A	N/A
7206107	12/2006	Levola	N/A	N/A
7212175	12/2006	Magee et al.	N/A	N/A
7218817	12/2006	Magnusson et al.	N/A	N/A
7230767	12/2006	Walck et al.	N/A	N/A
7230770	12/2006	Kreitzer et al.	N/A	N/A
7242527	12/2006	Spitzer et al.	N/A	N/A
7248128	12/2006	Mattila et al.	N/A	N/A
7248765	12/2006	Lee et al.	N/A	N/A
7256915	12/2006	Sutherland et al.	N/A	N/A
7259906	12/2006	Islam	N/A	N/A
7265882	12/2006	Sutherland et al.	N/A	N/A
7265903	12/2006	Sutherland et al.	N/A	N/A
7268946	12/2006	Wang	N/A	N/A
7280722	12/2006	Temkin et al.	N/A	N/A
7285903	12/2006	Cull et al.	N/A	N/A
7286272	12/2006	Mukawa	N/A	N/A
7289069	12/2006	Ranta	N/A	N/A
RE39911	12/2006	Moskovich	N/A	N/A
7299983	12/2006	Piikivi	N/A	N/A
7301601	12/2006	Lin et al.	N/A	N/A
7312906	12/2006	Sutherland et al.	N/A	N/A
7313291	12/2006	Okhotnikov et al.	N/A	N/A
D559250	12/2007	Pombo	N/A	N/A
7319573	12/2007	Nishiyama	N/A	N/A
7320534	12/2007	Sugikawa et al.	N/A	N/A
7323275	12/2007	Otaki et al.	N/A	N/A
7333685	12/2007	Stone et al.	N/A	N/A
7336271	12/2007	Ozeki et al.	N/A	N/A
7339737	12/2007	Urey et al.	N/A	N/A
7339742	12/2007	Amitai et al.	N/A	N/A
7349612	12/2007	Nishii et al.	N/A	N/A
7356218	12/2007	Kato et al.	N/A	N/A
7356224	12/2007	Levner et al.	N/A	N/A
7369911	12/2007	Volant et al.	N/A	N/A
7375870	12/2007	Schorpp	N/A	N/A
7375886	12/2007	Lipton et al.	N/A	N/A

7376068	12/2007	Khoury	N/A	N/A
7376307	12/2007	Singh et al.	N/A	N/A
7389023	12/2007	Yeo et al.	N/A	N/A
7391573	12/2007	Amitai	N/A	N/A
7394865	12/2007	Borran et al.	N/A	N/A
7394961	12/2007	Kornilovich et al.	N/A	N/A
7395181	12/2007	Foxlin	N/A	N/A
7397606	12/2007	Peng et al.	N/A	N/A
7401920	12/2007	Kranz et al.	N/A	N/A
7404644	12/2007	Evans et al.	N/A	N/A
7410286	12/2007	Travis	N/A	N/A
7411637	12/2007	Weiss	N/A	N/A
7413678	12/2007	Natarajan et al.	N/A	N/A
7413679	12/2007	Sutherland et al.	N/A	N/A
7415173	12/2007	Kassamakov et al.	N/A	N/A
7416818	12/2007	Sutherland et al.	N/A	N/A
7418170	12/2007	Mukawa et al.	N/A	N/A
7420733	12/2007	Natarajan et al.	N/A	N/A
7433116	12/2007	Islam	N/A	N/A
7436568	12/2007	Kuykendall, Jr.	N/A	N/A
D581447	12/2007	Yee	N/A	N/A
7447967	12/2007	Onggosanusi et al.	N/A	N/A
7453612	12/2007	Mukawa	N/A	N/A
7454103	12/2007	Parriaux	N/A	N/A
7457040	12/2007	Amitai	N/A	N/A
7466994	12/2007	Pihlaja et al.	N/A	N/A
7477206	12/2008	Cowan et al.	N/A	N/A
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7480215	12/2008	Makela et al.	N/A	N/A
7482996	12/2008	Larson et al.	N/A	N/A
7483604	12/2008	Levola	N/A	N/A
7492512	12/2008	Niv et al.	N/A	N/A
7496293	12/2008	Shamir et al.	N/A	N/A
7499217	12/2008	Cakmakci et al.	N/A	N/A
7500104	12/2008	Goland	N/A	N/A
7511891	12/2008	Messerschmidt	N/A	N/A
7513668	12/2008	Peng et al.	N/A	N/A
7522344	12/2008	Curatu et al.	N/A	N/A
7525448	12/2008	Wilson et al.	N/A	N/A
7528385	12/2008	Volodin et al.	N/A	N/A
7542210	12/2008	Chirieleison	N/A	N/A
7545429	12/2008	Travis	N/A	N/A
7550234	12/2008	Otaki et al.	N/A	N/A
7558446	12/2008	Wimberger-friedl et al.	N/A	N/A
7567372	12/2008	Schorpp	N/A	N/A
7570322	12/2008	Sutherland et al.	N/A	N/A
7570405	12/2008	Sutherland et al.	N/A	N/A
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7572555	12/2008	Takizawa et al.	N/A	N/A

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7576916	12/2008	Amitai	N/A	N/A
7577326	12/2008	Amitai	N/A	N/A
7579119	12/2008	Ueda et al.	N/A	N/A
7583423	12/2008	Sutherland et al.	N/A	N/A
7587110	12/2008	Singh et al.	N/A	N/A
7588863	12/2008	Takizawa et al.	N/A	N/A
7589900	12/2008	Powell	N/A	N/A
7589901	12/2008	DeJong et al.	N/A	N/A
7592988	12/2008	Katase	N/A	N/A
7593575	12/2008	Houle et al.	N/A	N/A
7597447	12/2008	Larson et al.	N/A	N/A
7599012	12/2008	Nakamura et al.	N/A	N/A
7600893	12/2008	Laino et al.	N/A	N/A
7602552	12/2008	Blumenfeld	N/A	N/A
7605719	12/2008	Wenger et al.	N/A	N/A
7605774	12/2008	Brandt et al.	N/A	N/A
7605882	12/2008	Sutherland et al.	N/A	N/A
7616270	12/2008	Hirabayashi et al.	N/A	N/A
7617022	12/2008	Wood et al.	N/A	N/A
7618750	12/2008	Ueda et al.	N/A	N/A
7619739	12/2008	Sutherland et al.	N/A	N/A
7619825	12/2008	Peng et al.	N/A	N/A
7629086	12/2008	Otaki et al.	N/A	N/A
7639208	12/2008	Ha et al.	N/A	N/A
7639911	12/2008	Lee et al.	N/A	N/A
7643214	12/2009	Amitai	N/A	N/A
7643225	12/2009	Tsai	N/A	N/A
7656585	12/2009	Powell et al.	N/A	N/A
7660047	12/2009	Travis et al.	N/A	N/A
7672024	12/2009	Kuan	N/A	N/A
7672055	12/2009	Amitai	N/A	N/A
7672549	12/2009	Ghosh et al.	N/A	N/A
7675021	12/2009	Lapstun	N/A	N/A
7675684	12/2009	Weissman et al.	N/A	N/A
7691248	12/2009	Ikeda et al.	N/A	N/A
7710622	12/2009	Takabayashi et al.	N/A	N/A
7710654	12/2009	Ashkenazi et al.	N/A	N/A
7711228	12/2009	Noda et al.	N/A	N/A
7724441	12/2009	Amitai	N/A	N/A
7724442	12/2009	Amitai	N/A	N/A
7724443	12/2009	Amitai	N/A	N/A
7733571	12/2009	Li	N/A	N/A
7733572	12/2009	Brown et al.	N/A	N/A
7740387	12/2009	Schultz et al.	N/A	N/A
7747113	12/2009	Mukawa et al.	N/A	N/A
7751122	12/2009	Amitai	N/A	N/A
7751662	12/2009	Kleemann et al.	N/A	N/A
7764413	12/2009	Levola	N/A	N/A
7777819	12/2009	Simmonds	N/A	N/A

7778305	12/2009	Parriaux et al.	N/A	N/A
7778508	12/2009	Hirayama	N/A	N/A
7843642	12/2009	Shaoulov et al.	N/A	N/A
7847235	12/2009	Krupkin et al.	N/A	N/A
7864427	12/2010	Korenaga et al.	N/A	N/A
7865080	12/2010	Hecker et al.	N/A	N/A
7866869	12/2010	Karakawa	N/A	N/A
7872707	12/2010	Sutherland et al.	N/A	N/A
7872804	12/2010	Moon et al.	N/A	N/A
7884593	12/2010	Simmonds et al.	N/A	N/A
7884985	12/2010	Amitai et al.	N/A	N/A
7887186	12/2010	Watanabe	N/A	N/A
7903921	12/2010	Ostergard	N/A	N/A
7907342	12/2010	Simmonds et al.	N/A	N/A
7920787	12/2010	Gentner et al.	N/A	N/A
7928862	12/2010	Matthews	N/A	N/A
7936513	12/2010	Wu et al.	N/A	N/A
7936519	12/2010	Mukawa et al.	N/A	N/A
7944428	12/2010	Travis	N/A	N/A
7944616	12/2010	Mukawa	N/A	N/A
7949214	12/2010	DeJong et al.	N/A	N/A
D640310	12/2010	Suzuki et al.	N/A	N/A
7961117	12/2010	Zimmerman et al.	N/A	N/A
7969644	12/2010	Tilleman et al.	N/A	N/A
7969657	12/2010	Cakmakci et al.	N/A	N/A
7970246	12/2010	Travis et al.	N/A	N/A
7976208	12/2010	Travis	N/A	N/A
7984884	12/2010	Iliev et al.	N/A	N/A
7999982	12/2010	Endo et al.	N/A	N/A
8000020	12/2010	Amitai et al.	N/A	N/A
8000491	12/2010	Brodkin et al.	N/A	N/A
8004765	12/2010	Amitai	N/A	N/A
8014050	12/2010	McGrew	N/A	N/A
8016475	12/2010	Travis	N/A	N/A
8018579	12/2010	Krah	N/A	N/A
8022942	12/2010	Bathiche et al.	N/A	N/A
8023783	12/2010	Mukawa et al.	N/A	N/A
RE42992	12/2010	David	N/A	N/A
8073296	12/2010	Mukawa et al.	N/A	N/A
8077274	12/2010	Sutherland et al.	N/A	N/A
8079713	12/2010	Ashkenazi	N/A	N/A
8082222	12/2010	Rangarajan et al.	N/A	N/A
8086030	12/2010	Gordon et al.	N/A	N/A
8089568	12/2011	Brown et al.	N/A	N/A
8093451	12/2011	Spangenberg et al.	N/A	N/A
8098439	12/2011	Amitai et al.	N/A	N/A
8105662	12/2011	Cherkaoui et al.	N/A	N/A
8107023	12/2011	Simmonds et al.	N/A	N/A
8107780	12/2011	Simmonds	N/A	N/A
8120548	12/2011	Barber	N/A	N/A

8120848	12/2011	Isano	N/A	N/A
8132948	12/2011	Owen et al.	N/A	N/A
8132976	12/2011	Odell et al.	N/A	N/A
8134434	12/2011	Diederichs et al.	N/A	N/A
8136690	12/2011	Fang et al.	N/A	N/A
8137981	12/2011	Andrew et al.	N/A	N/A
8142016	12/2011	Legerton et al.	N/A	N/A
8149086	12/2011	Klein et al.	N/A	N/A
8152315	12/2011	Travis et al.	N/A	N/A
8152353	12/2011	Yang et al.	N/A	N/A
8155489	12/2011	Saarikko et al.	N/A	N/A
8159752	12/2011	Wertheim et al.	N/A	N/A
8160409	12/2011	Large	N/A	N/A
8160411	12/2011	Levola et al.	N/A	N/A
D659137	12/2011	Matsumoto	N/A	N/A
8167173	12/2011	Simmonds et al.	N/A	N/A
8186874	12/2011	Sinbar et al.	N/A	N/A
8188925	12/2011	DeJean	N/A	N/A
8189263	12/2011	Wang et al.	N/A	N/A
8189973	12/2011	Travis et al.	N/A	N/A
D661334	12/2011	Cho et al.	N/A	N/A
D661335	12/2011	Jeon	N/A	N/A
8194325	12/2011	Levola et al.	N/A	N/A
8199803	12/2011	Hauske et al.	N/A	N/A
8202405	12/2011	Meneghini et al.	N/A	N/A
8213065	12/2011	Mukawa	N/A	N/A
8213755	12/2011	Mukawa et al.	N/A	N/A
8220966	12/2011	Mukawa	N/A	N/A
8224133	12/2011	Popovich et al.	N/A	N/A
8233204	12/2011	Robbins et al.	N/A	N/A
8253914	12/2011	Kajiya et al.	N/A	N/A
8254031	12/2011	Levola	N/A	N/A
8264498	12/2011	Vanderkamp et al.	N/A	N/A
8294749	12/2011	Cable	N/A	N/A
8295710	12/2011	Marcus	N/A	N/A
8301031	12/2011	Gentner et al.	N/A	N/A
8305577	12/2011	Kivioja et al.	N/A	N/A
8306423	12/2011	Gottwald et al.	N/A	N/A
8310327	12/2011	Willers et al.	N/A	N/A
8314819	12/2011	Kimmel et al.	N/A	N/A
8314993	12/2011	Levola et al.	N/A	N/A
8320032	12/2011	Levola	N/A	N/A
8321810	12/2011	Heintze	N/A	N/A
8325166	12/2011	Akutsu et al.	N/A	N/A
8329773	12/2011	Fäcke et al.	N/A	N/A
8335040	12/2011	Mukawa et al.	N/A	N/A
8335414	12/2011	Zinoviev et al.	N/A	N/A
D673996	12/2012	Kim et al.	N/A	N/A
8351744	12/2012	Travis et al.	N/A	N/A
8354640	12/2012	Hamre et al.	N/A	N/A

8354806	12/2012	Travis et al.	N/A	N/A
8355610	12/2012	Simmonds	N/A	N/A
8369019	12/2012	Baker et al.	N/A	N/A
8376548	12/2012	Schultz	N/A	N/A
8382293	12/2012	Phillips et al.	N/A	N/A
8384504	12/2012	Diederichs et al.	N/A	N/A
8384694	12/2012	Powell et al.	N/A	N/A
8384730	12/2012	Vanderkamp et al.	N/A	N/A
8396339	12/2012	Mukawa et al.	N/A	N/A
8396341	12/2012	Lee et al.	N/A	N/A
8398242	12/2012	Yamamoto et al.	N/A	N/A
8403490	12/2012	Sugiyama et al.	N/A	N/A
8422840	12/2012	Large	N/A	N/A
8427439	12/2012	Larsen et al.	N/A	N/A
8432363	12/2012	Saarikko et al.	N/A	N/A
8432372	12/2012	Butler et al.	N/A	N/A
8432614	12/2012	Amitai	N/A	N/A
8441731	12/2012	Sprague	N/A	N/A
8447365	12/2012	Immanuel	N/A	N/A
8466953	12/2012	Levola	N/A	N/A
8472119	12/2012	Kelly	N/A	N/A
8472120	12/2012	Border et al.	N/A	N/A
8477261	12/2012	Travis et al.	N/A	N/A
8481130	12/2012	Harding et al.	N/A	N/A
8482858	12/2012	Sprague	N/A	N/A
8488246	12/2012	Border et al.	N/A	N/A
8491121	12/2012	Tilleman et al.	N/A	N/A
8491136	12/2012	Travis et al.	N/A	N/A
8493366	12/2012	Bathiche et al.	N/A	N/A
8493562	12/2012	Kopp et al.	N/A	N/A
8493662	12/2012	Noui	N/A	N/A
8494229	12/2012	Jarvenpaa et al.	N/A	N/A
8508848	12/2012	Saarikko	N/A	N/A
8520309	12/2012	Sprague	N/A	N/A
D691192	12/2012	Stanley et al.	N/A	N/A
8547638	12/2012	Levola	N/A	N/A
8548290	12/2012	Travers et al.	N/A	N/A
8565560	12/2012	Popovich et al.	N/A	N/A
D694310	12/2012	Cho et al.	N/A	N/A
D694311	12/2012	Cho et al.	N/A	N/A
8578038	12/2012	Kaikuranta et al.	N/A	N/A
8581831	12/2012	Travis	N/A	N/A
8582206	12/2012	Travis	N/A	N/A
8593734	12/2012	Laakkonen	N/A	N/A
8611014	12/2012	Valera et al.	N/A	N/A
8619062	12/2012	Powell et al.	N/A	N/A
D697130	12/2013	Lövgren	N/A	N/A
8633786	12/2013	Ermolov et al.	N/A	N/A
8634120	12/2013	Popovich et al.	N/A	N/A
8634139	12/2013	Brown et al.	N/A	N/A

8639072	12/2013	Popovich et al.	N/A	N/A
8643691	12/2013	Rosenfeld et al.	N/A	N/A
8643948	12/2013	Amitai et al.	N/A	N/A
8649099	12/2013	Schultz et al.	N/A	N/A
8654420	12/2013	Simmonds	N/A	N/A
8659826	12/2013	Brown et al.	N/A	N/A
D701206	12/2013	Luckey et al.	N/A	N/A
8670029	12/2013	McEldowney	N/A	N/A
8693087	12/2013	Nowatzyk et al.	N/A	N/A
8698705	12/2013	Burke	N/A	N/A
8731350	12/2013	Lin et al.	N/A	N/A
8736802	12/2013	Kajiya et al.	N/A	N/A
8736963	12/2013	Robbins et al.	N/A	N/A
8742952	12/2013	Bold	N/A	N/A
8746008	12/2013	Mauritsen et al.	N/A	N/A
8749886	12/2013	Gupta	N/A	N/A
8749890	12/2013	Wood et al.	N/A	N/A
8767294	12/2013	Chen et al.	N/A	N/A
8786923	12/2013	Chuang et al.	N/A	N/A
8810600	12/2013	Bohn et al.	N/A	N/A
8810913	12/2013	Simmonds et al.	N/A	N/A
8810914	12/2013	Amitai	N/A	N/A
8814691	12/2013	Haddick et al.	N/A	N/A
8816578	12/2013	Peng et al.	N/A	N/A
8817350	12/2013	Robbins et al.	N/A	N/A
8824836	12/2013	Sugiyama	N/A	N/A
8830143	12/2013	Pitchford et al.	N/A	N/A
8830584	12/2013	Saarikko et al.	N/A	N/A
8830588	12/2013	Brown et al.	N/A	N/A
8842368	12/2013	Simmonds et al.	N/A	N/A
8859412	12/2013	Jain	N/A	N/A
8872435	12/2013	Kreitzer et al.	N/A	N/A
8873149	12/2013	Bohn et al.	N/A	N/A
8873150	12/2013	Amitai	N/A	N/A
D718304	12/2013	Heinrich	N/A	N/A
D718366	12/2013	Mehin et al.	N/A	N/A
8885112	12/2013	Popovich et al.	N/A	N/A
8885997	12/2013	Nguyen et al.	N/A	N/A
8903207	12/2013	Brown et al.	N/A	N/A
8906088	12/2013	Pugh et al.	N/A	N/A
8913324	12/2013	Schrader	N/A	N/A
8913865	12/2013	Bennett	N/A	N/A
8917453	12/2013	Bohn	N/A	N/A
8917962	12/2013	Nichol et al.	N/A	N/A
8929589	12/2014	Publicover et al.	N/A	N/A
8933144	12/2014	Enomoto et al.	N/A	N/A
8934743	12/2014	Nishiwaki et al.	N/A	N/A
8937771	12/2014	Robbins et al.	N/A	N/A
8937772	12/2014	Burns et al.	N/A	N/A
8938141	12/2014	Magnusson	N/A	N/A

8950867	12/2014	Macnamara	N/A	N/A
8964298	12/2014	Haddick et al.	N/A	N/A
8965152	12/2014	Simmonds	N/A	N/A
D725102	12/2014	Lee et al.	N/A	N/A
8985803	12/2014	Bohn	N/A	N/A
8989535	12/2014	Robbins	N/A	N/A
D726180	12/2014	Roat et al.	N/A	N/A
9019595	12/2014	Jain	N/A	N/A
9025253	12/2014	Hadad et al.	N/A	N/A
9035344	12/2014	Jain	N/A	N/A
D733709	12/2014	Kawai	N/A	N/A
9075184	12/2014	Popovich et al.	N/A	N/A
9081178	12/2014	Simmonds et al.	N/A	N/A
9097890	12/2014	Miller et al.	N/A	N/A
9103978	12/2014	Nishiwaki et al.	N/A	N/A
9122015	12/2014	Shimizu	N/A	N/A
9128226	12/2014	Fattal et al.	N/A	N/A
9129295	12/2014	Border et al.	N/A	N/A
9164290	12/2014	Robbins et al.	N/A	N/A
9176324	12/2014	Scherer et al.	N/A	N/A
9188717	12/2014	Nishiwaki	N/A	N/A
9201270	12/2014	Fattal et al.	N/A	N/A
9215293	12/2014	Miller	N/A	N/A
D746896	12/2015	Markovitz et al.	N/A	N/A
9239507	12/2015	Chen et al.	N/A	N/A
9244275	12/2015	Li	N/A	N/A
9244280	12/2015	Tiana et al.	N/A	N/A
9244281	12/2015	Zimmerman et al.	N/A	N/A
D749074	12/2015	Cazalet et al.	N/A	N/A
9253359	12/2015	Takahashi	N/A	N/A
9269854	12/2015	Jain	N/A	N/A
D751551	12/2015	Ho et al.	N/A	N/A
D752129	12/2015	Lee et al.	N/A	N/A
9274338	12/2015	Robbins et al.	N/A	N/A
9274339	12/2015	Brown et al.	N/A	N/A
9274349	12/2015	Popovich et al.	N/A	N/A
D754782	12/2015	Kokinakis et al.	N/A	N/A
9310566	12/2015	Valera et al.	N/A	N/A
9316786	12/2015	Nishiwaki et al.	N/A	N/A
9329325	12/2015	Simmonds et al.	N/A	N/A
9335548	12/2015	Cakmakci et al.	N/A	N/A
9335604	12/2015	Popovich et al.	N/A	N/A
9341846	12/2015	Popovich et al.	N/A	N/A
9354366	12/2015	Jain	N/A	N/A
9366862	12/2015	Haddick et al.	N/A	N/A
9366864	12/2015	Brown et al.	N/A	N/A
9372347	12/2015	Levola et al.	N/A	N/A
9377623	12/2015	Robbins et al.	N/A	N/A
9377852	12/2015	Shapiro et al.	N/A	N/A
9389415	12/2015	Fattal et al.	N/A	N/A

9400395	12/2015	Travers et al.	N/A	N/A
9423360	12/2015	Kostamo et al.	N/A	N/A
9429692	12/2015	Saarikko et al.	N/A	N/A
9431794	12/2015	Jain	N/A	N/A
9435961	12/2015	Jiang	N/A	N/A
9456744	12/2015	Popovich et al.	N/A	N/A
9459451	12/2015	Saarikko et al.	N/A	N/A
9464779	12/2015	Popovich et al.	N/A	N/A
9465213	12/2015	Simmonds	N/A	N/A
9465227	12/2015	Popovich et al.	N/A	N/A
9484482	12/2015	Hsu et al.	N/A	N/A
9494799	12/2015	Robbins et al.	N/A	N/A
9507150	12/2015	Stratton et al.	N/A	N/A
9513480	12/2015	Saarikko et al.	N/A	N/A
9516193	12/2015	Aramaki	N/A	N/A
9519089	12/2015	Brown et al.	N/A	N/A
9519115	12/2015	Yashiki et al.	N/A	N/A
9523852	12/2015	Brown et al.	N/A	N/A
9535253	12/2016	Levola et al.	N/A	N/A
9541383	12/2016	Abovitz et al.	N/A	N/A
9541763	12/2016	Heberlein et al.	N/A	N/A
9547174	12/2016	Gao et al.	N/A	N/A
9551468	12/2016	Jones	N/A	N/A
9551874	12/2016	Amitai	N/A	N/A
9551880	12/2016	Amitai	N/A	N/A
9599813	12/2016	Stratton et al.	N/A	N/A
9612403	12/2016	Abovitz et al.	N/A	N/A
9632226	12/2016	Waldern et al.	N/A	N/A
9635352	12/2016	Henry et al.	N/A	N/A
9648313	12/2016	Henry et al.	N/A	N/A
9651368	12/2016	Abovitz et al.	N/A	N/A
9664824	12/2016	Simmonds et al.	N/A	N/A
9664910	12/2016	Mansharof et al.	N/A	N/A
9671612	12/2016	Kress et al.	N/A	N/A
9674413	12/2016	Tiana et al.	N/A	N/A
9678345	12/2016	Melzer et al.	N/A	N/A
9679367	12/2016	Wald	N/A	N/A
9715067	12/2016	Brown et al.	N/A	N/A
9715110	12/2016	Brown et al.	N/A	N/A
D793468	12/2016	Yu et al.	N/A	N/A
D795865	12/2016	Porter et al.	N/A	N/A
D795866	12/2016	Porter et al.	N/A	N/A
9726540	12/2016	Popovich et al.	N/A	N/A
9727772	12/2016	Popovich et al.	N/A	N/A
9733475	12/2016	Brown et al.	N/A	N/A
9739950	12/2016	Sqalli et al.	N/A	N/A
9746688	12/2016	Popovich et al.	N/A	N/A
9754507	12/2016	Wenger et al.	N/A	N/A
9762895	12/2016	Henry et al.	N/A	N/A
9766465	12/2016	Tiana et al.	N/A	N/A

9785231	12/2016	Zimmerman	N/A	N/A
9791694	12/2016	Haverkamp et al.	N/A	N/A
9791696	12/2016	Woltman et al.	N/A	N/A
9791703	12/2016	Vallius et al.	N/A	N/A
9804316	12/2016	Drolet et al.	N/A	N/A
9804389	12/2016	Popovich et al.	N/A	N/A
9823423	12/2016	Waldern et al.	N/A	N/A
9857605	12/2017	Popovich et al.	N/A	N/A
9874931	12/2017	Koenck et al.	N/A	N/A
9891436	12/2017	Wall et al.	N/A	N/A
9899800	12/2017	Ferrotti et al.	N/A	N/A
9915825	12/2017	Robbins et al.	N/A	N/A
9933684	12/2017	Brown et al.	N/A	N/A
9939577	12/2017	Inoue et al.	N/A	N/A
9939628	12/2017	Basset et al.	N/A	N/A
9959818	12/2017	Bohn	N/A	N/A
9977247	12/2017	Brown et al.	N/A	N/A
9989763	12/2017	Woltman et al.	N/A	N/A
10025093	12/2017	Wall et al.	N/A	N/A
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10067347	12/2017	Vallius et al.	N/A	N/A
10088675	12/2017	Brown et al.	N/A	N/A
10088686	12/2017	Robbins et al.	N/A	N/A
10089516	12/2017	Popovich et al.	N/A	N/A
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10107966	12/2017	Horibe et al.	N/A	N/A
10114220	12/2017	Grey et al.	N/A	N/A
10126552	12/2017	Brown et al.	N/A	N/A
10156681	12/2017	Waldern et al.	N/A	N/A
10162181	12/2017	Webster et al.	N/A	N/A
10185154	12/2018	Popovich et al.	N/A	N/A
D840454	12/2018	Han et al.	N/A	N/A
10197804	12/2018	Stenberg et al.	N/A	N/A
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10216061	12/2018	Popovich et al.	N/A	N/A
10234696	12/2018	Popovich et al.	N/A	N/A
10241330	12/2018	Popovich et al.	N/A	N/A
10241332	12/2018	Vallius	N/A	N/A
10247943	12/2018	Yu et al.	N/A	N/A
10281725	12/2018	Yokoyama	N/A	N/A
10330777	12/2018	Popovich et al.	N/A	N/A
10345519	12/2018	Miller et al.	N/A	N/A
10359627	12/2018	Wall et al.	N/A	N/A
10359635	12/2018	Grey et al.	N/A	N/A
10359736	12/2018	Popovich et al.	N/A	N/A
D855687	12/2018	Villalpando	N/A	N/A
D859510	12/2018	Harmon et al.	N/A	N/A
10409144	12/2018	Popovich et al.	N/A	N/A
10423222	12/2018	Popovich et al.	N/A	N/A
10423813	12/2018	Popovich et al.	N/A	N/A

10437051	12/2018	Popovich et al.	N/A	N/A
10437064	12/2018	Popovich et al.	N/A	N/A
10444510	12/2018	Lee et al.	N/A	N/A
10459145	12/2018	Popovich et al.	N/A	N/A
10459311	12/2018	Popovich et al.	N/A	N/A
D871494	12/2018	Yamada et al.	N/A	N/A
10509241	12/2018	Robbins et al.	N/A	N/A
D872170	12/2019	Evans et al.	N/A	N/A
D872794	12/2019	Wilkins	N/A	N/A
10527797	12/2019	Waldern et al.	N/A	N/A
10532594	12/2019	Akahane et al.	N/A	N/A
10545346	12/2019	Waldern et al.	N/A	N/A
10551616	12/2019	Wall et al.	N/A	N/A
10560688	12/2019	Robbins	N/A	N/A
10569449	12/2019	Curts et al.	N/A	N/A
10578876	12/2019	Lam et al.	N/A	N/A
10591756	12/2019	Popovich et al.	N/A	N/A
10598938	12/2019	Huang et al.	N/A	N/A
D880575	12/2019	Thixton	N/A	N/A
10613268	12/2019	Colburn et al.	N/A	N/A
10642058	12/2019	Popovich et al.	N/A	N/A
10649119	12/2019	Mohanty et al.	N/A	N/A
10670876	12/2019	Popovich et al.	N/A	N/A
10678053	12/2019	Waldern et al.	N/A	N/A
10690831	12/2019	Calafiore	N/A	N/A
10690851	12/2019	Waldern et al.	N/A	N/A
10690915	12/2019	Popovich et al.	N/A	N/A
10690916	12/2019	Popovich et al.	N/A	N/A
10698214	12/2019	Vallius et al.	N/A	N/A
10705281	12/2019	Fattal et al.	N/A	N/A
10712571	12/2019	Popovich et al.	N/A	N/A
10725312	12/2019	Popovich et al.	N/A	N/A
10732351	12/2019	Colburn et al.	N/A	N/A
10732569	12/2019	Waldern et al.	N/A	N/A
10746989	12/2019	Brown et al.	N/A	N/A
10747982	12/2019	Popovich et al.	N/A	N/A
10795160	12/2019	Stanley et al.	N/A	N/A
10823887	12/2019	Calafiore et al.	N/A	N/A
10859768	12/2019	Popovich et al.	N/A	N/A
10859837	12/2019	Adema et al.	N/A	N/A
10890707	12/2020	Waldern et al.	N/A	N/A
10914950	12/2020	Waldern et al.	N/A	N/A
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10983257	12/2020	Colburn et al.	N/A	N/A
10983340	12/2020	Popovich et al.	N/A	N/A
10983346	12/2020	Vallius et al.	N/A	N/A
11009699	12/2020	Popovich et al.	N/A	N/A
11103892	12/2020	Liao et al.	N/A	N/A
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11137603	12/2020	Zhang	N/A	N/A
11175512	12/2020	Waldern et al.	N/A	N/A
11194098	12/2020	Waldern et al.	N/A	N/A
11194159	12/2020	Popovich et al.	N/A	N/A
11194162	12/2020	Waldern et al.	N/A	N/A
11204540	12/2020	Popovich et al.	N/A	N/A
11231544	12/2021	Lin et al.	N/A	N/A
11243333	12/2021	Ouderkirk et al.	N/A	N/A
11256155	12/2021	Popovich et al.	N/A	N/A
11281013	12/2021	Popovich et al.	N/A	N/A
11300795	12/2021	Stanley et al.	N/A	N/A
11306193	12/2021	Lane et al.	N/A	N/A
11307357	12/2021	Mohanty	N/A	N/A
11307432	12/2021	Popovich et al.	N/A	N/A
11320571	12/2021	Brown et al.	N/A	N/A
11340386	12/2021	Ouderkirk et al.	N/A	N/A
11378732	12/2021	Waldern et al.	N/A	N/A
11391950	12/2021	Calafiore	N/A	N/A
11402801	12/2021	Waldern et al.	N/A	N/A
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11448937	12/2021	Brown et al.	N/A	N/A
11513350	12/2021	Waldern et al.	N/A	N/A
11586046	12/2022	Waldern et al.	N/A	N/A
11604314	12/2022	Popovich et al.	N/A	N/A
11703645	12/2022	Waldern et al.	N/A	N/A
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2009/0097127	12/2008	Amitai	N/A	N/A
2009/0116790	12/2008	Mossberg et al.	N/A	N/A
2009/0121301	12/2008	Chang	N/A	N/A
2009/0122413	12/2008	Hoffman et al.	N/A	N/A
2009/0122414	12/2008	Amitai	N/A	N/A
2009/0128495	12/2008	Kong et al.	N/A	N/A
2009/0128781	12/2008	Li	N/A	N/A
2009/0128902	12/2008	Niv et al.	N/A	N/A
2009/0128911	12/2008	Itzkovitch et al.	N/A	N/A
2009/0136246	12/2008	Murakami	N/A	N/A
2009/0141324	12/2008	Mukawa	N/A	N/A
2009/0153437	12/2008	Aharoni	N/A	N/A
2009/0169152	12/2008	Oestergard	N/A	N/A
2009/0190222	12/2008	Simmonds et al.	N/A	N/A
2009/0213208	12/2008	Glatt	N/A	N/A
2009/0237804	12/2008	Amitai et al.	N/A	N/A
2009/0242021	12/2008	Petkie et al.	N/A	N/A
2009/0296218	12/2008	Ryytty	N/A	N/A
2009/0303599	12/2008	Levola	N/A	N/A
2009/0316246	12/2008	Asai et al.	N/A	N/A
2010/0014312	12/2009	Travis et al.	N/A	N/A
2010/0039796	12/2009	Mukawa	N/A	N/A
2010/0053565	12/2009	Mizushima et al.	N/A	N/A
2010/0060551	12/2009	Sugiyama et al.	N/A	N/A
2010/0060990	12/2009	Wertheim et al.	N/A	N/A
2010/0065726	12/2009	Zhong et al.	N/A	N/A
2010/0079841	12/2009	Levola	N/A	N/A
2010/0079865	12/2009	Saarikko et al.	N/A	N/A
2010/0084261	12/2009	Lee et al.	N/A	N/A
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2010/0141555	12/2009	Rorberg et al.	N/A	N/A
2010/0141905	12/2009	Burke	N/A	N/A
2010/0149073	12/2009	Chaum et al.	N/A	N/A
2010/0165465	12/2009	Levola	N/A	N/A
2010/0165660	12/2009	Weber et al.	N/A	N/A

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2010/0177388	12/2009	Cohen et al.	N/A	N/A
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2010/0214659	12/2009	Levola	N/A	N/A
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2010/0225834	12/2009	Li	N/A	N/A
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2010/0231693	12/2009	Levola	N/A	N/A
2010/0231705	12/2009	Yahav et al.	N/A	N/A
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2010/0245756	12/2009	Sugihara et al.	N/A	N/A
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2010/0253987	12/2009	Leopold et al.	N/A	N/A
2010/0260030	12/2009	Yuyama et al.	N/A	N/A
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2011/0232211	12/2010	Farahi	N/A	N/A
2011/0235179	12/2010	Simmonds	N/A	N/A
2011/0235365	12/2010	McCollum et al.	N/A	N/A
2011/0236803	12/2010	Weiser et al.	N/A	N/A
2011/0238399	12/2010	Ophir et al.	N/A	N/A
2011/0242349	12/2010	Izuha et al.	N/A	N/A
2011/0242661	12/2010	Simmonds	N/A	N/A
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2013/0308185	12/2012	Robinson et al.	N/A	N/A
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2013/0314793	12/2012	Robbins et al.	N/A	N/A
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2013/0328948	12/2012	Kunkel et al.	N/A	N/A
2013/0342525	12/2012	Benko et al.	N/A	N/A
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2014/0043672	12/2013	Clarke et al.	N/A	N/A
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2014/0055845	12/2013	Jain	N/A	N/A
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2014/0098010	12/2013	Travis	N/A	N/A
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2015/0288129	12/2014	Jain	N/A	N/A
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The current application is a continuation of U.S. patent application Ser. No. 16/783,019 entitled “Methods for Compensating for Optical Surface Nonuniformity,” filed Feb. 5, 2020, which application claims the benefit of and priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/801,528 entitled “Methods for Compensating for Optical Surface Nonuniformity,” filed Feb. 5, 2019. The disclosures of which are hereby incorporated by reference in their entirety.

FIELD OF THE INVENTION

(1) The present invention generally relates to methods for compensating for nonuniform surface topography features in substrates and, more specifically, in waveguide substrates.

BACKGROUND

(2) Surface flatness is a common requirement in the manufacture of various semiconductor devices, including but not limited to display backplanes. In holographic waveguides used for conveying image data, tight planarity tolerances are desirable to avoid image artifacts that can result from light path deviations caused by surface nonuniformities. Industry methods for planarization of semiconductor substrates include low pressure chemical vapor deposition (LPCVD), etch-back of a sacrificial layer (typically a photoresist), simultaneous etch and deposition, chemical mechanical polishing (CMP), reflow of SiO₂ doped with phosphorus and boron glass, and other processes. Applications of such methods cover the range from smooth partial planarization all the way to complete global planarization. Existing processes of relevance to the manufacture of holographic waveguides can include organic spin coating, molding using a planar surface, and other coating processes.

SUMMARY OF THE INVENTION

(3) Systems and methods for compensating for nonuniform surface topography features in accordance with various embodiments of the invention are illustrated. One embodiment includes a method for manufacturing waveguide cells, the method including providing a waveguide including first and second substrates and a layer of optical recording material, and applying a surface forming process to at least one external surface of the first and second substrates.

(4) In another embodiment, applying the surface forming process includes applying a forming material coating to the at least one external surface, providing a forming element having a forming surface, bringing the forming element in physical contact with the forming material coating, curing the forming material coating while it is in contact with the forming element, and releasing the forming material coating from the forming element.

(5) In a further embodiment, the surface forming process provides surface planarization the at least one external surface.

(6) In still another embodiment, applying the surface forming process further includes depositing at least one of a release layer or a hard coat layer.

(7) In a still further embodiment, curing the forming material coating includes applying UV curing radiation via a curing configuration selected from the group consisting of one or more UV sources distributed above the forming material coating, a fixture including UV sources spatially distributed around the waveguide, and one or more UV sources coupled into an internal reflection path within at least one of the forming element or the waveguide.

(8) In yet another embodiment, curing the forming material coating includes a curing configuration selected from the group consisting of applying radiation having more than one wavelength, applying UV radiation having more than one wavelength within the UV spectrum, applying a

thermal process, applying pressure, immersing the forming material coating in a vacuum, immersing the forming material coating in a gas, immersing the forming material coating in a liquid, applying radiation having a wavelength within the visible spectrum, applying radiation having a wavelength within the infrared spectrum, and exposing the forming material coating to a humid atmosphere.

(9) In a yet further embodiment, the forming material coating includes a material selected from the group consisting of photoresists, resins, polymers, thermosets, thermoplastic polymers, polyepoxies, polyamides, low viscosity planarization materials, materials with viscosity between 10 and 15 cps at 20° C., materials with viscosity below 2 cps at 20° C., and ethylene glycol diacrylate.

(10) In another additional embodiment, the release layer is a fluorocarbon silylating agent.

(11) In a further additional embodiment, the forming surface of the forming element has a surface characteristic for compensating for shrinkage of the forming material coating.

(12) In another embodiment again, the method further includes correcting a wedge characteristic of the forming material coating after completion of the surface forming process.

(13) In a further embodiment again, the first substrate is curved, and the forming element surface has a curvature matching the curvature of the first substrate.

(14) In still yet another embodiment, the forming surface of the forming element has a degree of planarization less than 98%.

(15) In a still yet further embodiment, the first substrate is a plastic.

(16) In still another additional embodiment, the substrates have a birefringence calculated to provide smoothing of non-uniformity after a predefined number of TIR beam reflections.

(17) In a still further additional embodiment, the method further includes employing an apparatus for monitoring at least one selected from the group of thickness, composition, and uniformity of the forming material during deposition.

(18) In still another embodiment again, providing the waveguide includes providing the first substrate, depositing the layer of optical recording material onto the first substrate using at least one deposition head, wherein the optical recording material deposited over a grating region is formulated to achieve a predefined grating characteristic selected from the group consisting of refractive index modulation, refractive index, birefringence, liquid crystal director alignment, grating layer thickness, and a spatial variation of the characteristic, providing the second substrate, placing the second substrate onto the deposited layer of optical recording material, and laminating the first substrate, the layer of optical recording material, and the second substrate.

(19) In a still further embodiment again, providing the waveguide further includes depositing at least one of release layer or a hard coat layer using the at least one deposition head.

(20) In yet another additional embodiment, the forming element is an optical flat, a refractive optical element, or an optical element with at least one optical surface providing optical power.

(21) In a yet further additional embodiment, the forming element is supported by a substrate providing an optical path from a light source for curing.

(22) In yet another embodiment again, the layer of optical recording material includes a polymer dispersed liquid crystal mixture.

(23) Additional embodiments and features are set forth in part in the description that follows, and in part will become apparent to those skilled in the art upon examination of the specification or may be learned by the practice of the invention. A further understanding of the nature and advantages of the present invention may be realized by reference to the remaining portions of the specification and the drawings, which forms a part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The description will be more fully understood with reference to the following figures and data graphs, which are presented as exemplary embodiments of the invention and should not be construed as a complete recitation of the scope of the invention.
- (2) FIG. 1 shows a flow diagram conceptually illustrating a method of providing a planarized waveguide in accordance with an embodiment of the invention.
- (3) FIGS. 2A-2E conceptually illustrate a series of process steps for planarizing a waveguide in accordance with an embodiment of the invention.
- (4) FIG. 3 conceptually illustrates a curing configuration utilizing a curing source disposed above a transparent forming element in accordance with an embodiment of the invention.
- (5) FIG. 4 conceptually illustrates a curing configuration utilizing a curing source optically coupled to a forming element in accordance with an embodiment of the invention.
- (6) FIG. 5 conceptually illustrates a curing configuration utilizing a curing source optically coupled to a waveguide in accordance with an embodiment of the invention.

DETAILED DESCRIPTION

(7) For the purposes of describing embodiments, some well-known features of optical technology known to those skilled in the art of optical design and visual displays have been omitted or simplified in order to not obscure the basic principles of the invention. Unless otherwise stated, the term “on-axis” in relation to a ray or a beam direction refers to propagation parallel to an axis normal to the surfaces of the optical components described in relation to the invention. In the following description the terms light, ray, beam, and direction may be used interchangeably and in association with each other to indicate the direction of propagation of electromagnetic radiation along rectilinear trajectories. The term light and illumination may be used in relation to the visible and infrared bands of the electromagnetic spectrum. Parts of the following description will be presented using terminology commonly employed by those skilled in the art of optical design. As used herein, the term grating may encompass a grating comprised of a set of gratings in some embodiments. For illustrative purposes, it is to be understood that the drawings are not drawn to scale unless stated otherwise.

(8) Planarization processes in accordance with various embodiments of the invention includes processes that can enable low cost products and lightweight designs. In many embodiments, a planarization process is implemented for planarizing plastic substrates for use in waveguides. Such processes can be utilized to mitigate or eliminate undesirable surface characteristics of the plastic substrates. Undesirable defects can include wedge shapes, bending of the substrate, and high total thickness variation. In some embodiments, the planarization process is compatible with deposition and printing processes that can be used in the manufacturing of holographic waveguides. Such processes are described in further detail in U.S. application Ser. No. 16/203,071 filed Nov. 28, 2018 entitled “Systems and Methods for Manufacturing Waveguide Cells,” the disclosure of which is hereby incorporated by reference in its entirety for all purposes. In some embodiments, a planarization process is implemented for curved waveguides, which can be integrated into various applications such as but not limited to helmet visors and car windshields. In such cases, the planarization process can compensate for deviations from the desired curvature of the waveguide. In several embodiments, a planarization process is implemented for correcting nonuniformity in waveguides, including curved waveguides. Although planarization processes in accordance with various embodiments of the invention apply to the compensation of surface nonuniformities of waveguide substrates of any geometry, to simplify the description of such embodiments, the discussions below will generally address applications directed at planarizing nominally flat waveguide substrates.

(9) In many embodiments, the planarization process is performed on the outer surfaces of a fabricated waveguide. In some embodiments, the planarization process is performed on a substrate

before it is used to fabricate a waveguide. The planarization process can include coating the faces of a waveguide or substrate with a deformable material (referred to as a forming material) and then bringing it into contact with an element (referred to as a forming element) having a forming surface with a desired surface profile, typically a surface with a high degree of flatness. The forming material can be cured, and the outer surfaces of the finished waveguide or substrate can achieve flatness specifications substantially equal to those of the forming surface. The process can be applied to one or both of the external surfaces of the waveguide. In the case of a curved waveguide or substrate, the process can include using a curved forming surface instead of a flat forming surface. In a number of embodiments, the curved forming surface has a curvature matching that of the waveguide or substrate. Waveguide structures, planarization processes, and planarization materials utilized for such processes are described below in further detail.

(10) Optical Waveguides and Grating Structures

(11) Optical structures recorded in waveguides can include many different types of optical elements, such as but not limited to diffraction gratings. In many embodiments, the grating implemented is a Bragg grating (also referred to as a volume grating). Bragg gratings can have high efficiency with little light being diffracted into higher orders. The relative amount of light in the diffracted and zero order can be varied by controlling the refractive index modulation of the grating, a property that can be used to make lossy waveguide gratings for extracting light over a large pupil.

(12) One class of gratings used in holographic waveguide devices is the Switchable Bragg Grating (SBG). SBGs can be fabricated by first placing a thin film of a mixture of photopolymerizable monomers and liquid crystal material between substrates, forming a waveguide cell. Waveguide cells are described in the sections below in further detail. The substrates can be made of various types of materials, such glass and plastics. In many cases, the substrates are in a parallel configuration. One or both substrates can support electrodes, typically transparent tin oxide films, for applying an electric field across the film. The grating structure in an SBG can be recorded in the liquid material (often referred to as the syrup) through photopolymerization-induced phase separation using interferential exposure with a spatially periodic intensity modulation. Factors such as but not limited to control of the irradiation intensity, component volume fractions of the materials in the mixture, and exposure temperature can determine the resulting grating morphology and performance. As can readily be appreciated, a wide variety of materials and mixtures can be used depending on the specific requirements of a given application. In many embodiments, HPDLC material is used. During the recording process, the monomers polymerize, and the mixture undergoes a phase separation. The LC molecules aggregate to form discrete or coalesced droplets that are periodically distributed in polymer networks on the scale of optical wavelengths. The alternating liquid crystal-rich and liquid crystal-depleted regions form the fringe planes of the grating, which can produce Bragg diffraction with a strong optical polarization resulting from the orientation ordering of the LC molecules in the droplets.

(13) The resulting volume phase grating can exhibit very high diffraction efficiency, which can be controlled by the magnitude of the electric field applied across the film. When an electric field is applied to the grating via transparent electrodes, the natural orientation of the LC droplets can change, causing the refractive index modulation of the fringes to lower and the hologram diffraction efficiency to drop to very low levels. Typically, the electrodes are configured such that the applied electric field will be perpendicular to the substrates. In a number of embodiments, the electrodes are fabricated from indium tin oxide (ITO). In the OFF state with no electric field applied, the extraordinary axis of the liquid crystals generally aligns normal to the fringes. The grating thus exhibits high refractive index modulation and high diffraction efficiency for P-polarized light. When an electric field is applied to the HPDLC, the grating switches to the ON state wherein the extraordinary axes of the liquid crystal molecules align parallel to the applied field and hence perpendicular to the substrate. In the ON state, the grating exhibits lower refractive

index modulation and lower diffraction efficiency for both S- and P-polarized light. Thus, the grating region no longer diffracts light. Each grating region can be divided into a multiplicity of grating elements such as for example a pixel matrix according to the function of the HPDLC device. Typically, the electrode on one substrate surface is uniform and continuous, while electrodes on the opposing substrate surface are patterned in accordance to the multiplicity of selectively switchable grating elements.

(14) Typically, the SBG elements are switched clear in 30 us with a longer relaxation time to switch ON. The diffraction efficiency of the device can be adjusted, by means of the applied voltage, over a continuous range. In many cases, the device exhibits near 100% efficiency with no voltage applied and essentially zero efficiency with a sufficiently high voltage applied. In certain types of HPDLC devices, magnetic fields can be used to control the LC orientation. In some HPDLC applications, phase separation of the LC material from the polymer can be accomplished to such a degree that no discernible droplet structure results. An SBG can also be used as a passive grating. In this mode, its chief benefit is a uniquely high refractive index modulation. SBGs can be used to provide transmission or reflection gratings for free space applications. SBGs can be implemented as waveguide devices in which the HPDLC forms either the waveguide core or an evanescently coupled layer in proximity to the waveguide. The substrates used to form the HPDLC cell provide a total internal reflection (TIR) light guiding structure. Light can be coupled out of the SBG when the switchable grating diffracts the light at an angle beyond the TIR condition.

(15) Waveguides can be constructed using a variety of processes, including but not limited to those described in U.S. application Ser. No. 16/203,071. In many embodiments, the waveguide is fabricated with a grating having a predefined grating characteristic. One manufacturing process capable of fabricating such waveguides includes a deposition technique where a layer of optical recording material is deposited onto a first substrate using at least one deposition head. In some embodiments, the optical recording material is a polymer dispersed liquid crystal mixture. In a number of embodiments, the optical recording material does not include liquid crystals. As can readily be appreciated, any type of recording material can be utilized as appropriate depending on the specific requirements of a given application. The optical recording material deposited over the grating region can be formulated to form a grating having a predefined grating characteristic. In a number of embodiments, the predefined grating characteristic is at least one of: refractive index modulation, refractive index, birefringence, liquid crystal director alignment, grating layer thickness, and combinations thereof. In further embodiments, the grating is formed with a spatially varying predefined grating characteristic. A second substrate can be placed on the deposited layer of optical recording material, and an exposure process can be performed to form gratings within the optical recording material layer.

(16) In some embodiments, LC can be extracted or evacuated from the SBG to provide a surface relief grating (SRG) that has properties very similar to a Bragg grating due to the depth of the SRG structure (which is much greater than that practically achievable using surface etching and other conventional processes commonly used to fabricate SRGs). The LC can be extracted using a variety of different methods, including but not limited to flushing with isopropyl alcohol and solvents. In many embodiments, one of the transparent substrates of the SBG is removed, and the LC is extracted. In further embodiments, the removed substrate is replaced. The SRG can be at least partially backfilled with a material of higher or lower refractive index. Such gratings offer scope for tailoring the efficiency, angular/spectral response, polarization, and other properties to suit various waveguide applications.

(17) Waveguides in accordance with various embodiments of the invention can include various grating configurations designed for specific purposes and functions. In many embodiments, the waveguide includes an input grating optically coupled to a light source, a fold grating for providing a first direction beam expansion, and an output grating for providing beam expansion in a second direction, which is typically orthogonal to the first direction, and beam extraction towards the

eyebow. As can readily be appreciated, the grating configuration implemented waveguide architectures can depend on the specific requirements of a given application. In some embodiments, the grating configuration includes multiple fold gratings. In several embodiments, the grating configuration includes an input grating and a second grating for performing beam expansion and beam extraction simultaneously. The second grating can include gratings of different prescriptions, for propagating different portions of the field of view, arranged in separate overlapping grating layers or multiplexed in a single grating layer. Furthermore, various types of gratings and waveguide architectures can also be utilized.

(18) In many embodiments, the waveguide can incorporate at least one of: angle multiplexed gratings, color multiplexed gratings, fold gratings, dual interaction gratings, rolled K-vector gratings, crossed fold gratings, tessellated gratings, chirped gratings, gratings with spatially varying refractive index modulation, gratings having spatially varying grating thickness, gratings having spatially varying average refractive index, gratings with spatially varying refractive index modulation tensors, and gratings having spatially varying average refractive index tensors. In some embodiments, the waveguide can incorporate at least one of: a half wave plate, a quarter wave plate, an anti-reflection coating, a beam splitting layer, an alignment layer, a photochromic back layer for glare reduction, and louvre films for glare reduction. In several embodiments, the waveguide can support gratings providing separate optical paths for different polarizations. In various embodiments, the waveguide can support gratings providing separate optical paths for different spectral bandwidths. In a number of embodiments, the gratings can be HPDLC gratings, switching gratings recorded in HPDLC (such as switchable Bragg Gratings), Bragg gratings recorded in holographic photopolymer, or surface relief gratings. In many embodiments, the waveguide operates in a monochrome band. In some embodiments, the waveguide operates in the green band. In several embodiments, waveguide layers operating in different spectral bands such as red, green, and blue (RGB) can be stacked to provide a three-layer waveguiding structure. In further embodiments, the layers are stacked with air gaps between the waveguide layers. In various embodiments, the waveguide layers operate in broader bands such as blue-green and green-red to provide two-waveguide layer solutions. In other embodiments, the gratings are color multiplexed to reduce the number of grating layers. Various types of gratings can be implemented. In some embodiments, at least one grating in each layer is a switchable grating.

(19) Waveguides incorporating optical structures such as those discussed above can be implemented in a variety of different applications, including but not limited to waveguide displays. In various embodiments, the waveguide display is implemented with an eyebow of greater than 10 mm with an eye relief greater than 25 mm. In some embodiments, the waveguide display includes a waveguide with a thickness between 2.0-5.0 mm. In many embodiments, the waveguide display can provide an image field of view of at least 50° diagonal. In further embodiments, the waveguide display can provide an image field of view of at least 70° diagonal. The waveguide display can employ many different types of picture generation units (PGUs). In several embodiments, the PGU can be a reflective or transmissive spatial light modulator such as a liquid crystal on Silicon (LCoS) panel or a micro electromechanical system (MEMS) panel. In a number of embodiments, the PGU can be an emissive device such as an organic light emitting diode (OLED) panel. In some embodiments, an OLED display can have a luminance greater than 4000 nits and a resolution of 4k×4k pixels. In several embodiments, the waveguide can have an optical efficiency greater than 10% such that a greater than 400 nit image luminance can be provided using an OLED display of luminance 4000 nits. Waveguides implementing P-diffracting gratings (i.e., gratings with high efficiency for P-polarized light) typically have a waveguide efficiency of 5%-6.2%. Since P-diffracting or S-diffracting gratings can waste half of the light from an unpolarized source such as an OLED panel, many embodiments are directed towards waveguides capable of providing both S-diffracting and P-diffracting gratings to allow for an increase in the efficiency of the waveguide by up to a factor of two. In some embodiments, the S-diffracting and P-diffracting gratings are

implemented in separate overlapping grating layers. Alternatively, a single grating can, under certain conditions, provide high efficiency for both p-polarized and s-polarized light. In several embodiments, the waveguide includes Bragg-like gratings produced by extracting LC from HPDLC gratings, such as those described above, to enable high S and P diffraction efficiency over certain wavelength and angle ranges for suitably chosen values of grating thickness (typically, in the range 2-5 μm).

(20) Planarization Processes

(21) Planarization processes in accordance with various embodiments of the invention can utilize a variety of different types of substrates, including but not limited to glass and plastic substrates of different thicknesses and geometries. Plastic substrates can include but are not limited to polyvinyl butyral (PVB), cyclo-olefinic polymers (COP), polymethyl methacrylate (PMMA), polycarbonates (PC), clear polyimides, etc. In many embodiments, the substrate can be injection molded. In some embodiments, the substrate has low birefringence. In several embodiments, the substrate has a birefringence calculated to provide smoothing of illumination non-uniformities occurring in the finished waveguide in normal operation. The planarization process can be performed on individual substrates or the outer surfaces of constructed waveguides. FIG. 1 shows a flow diagram conceptually illustrating a method of providing a planarized waveguide in accordance with an embodiment of the invention. Referring to the flow diagram, the method **100** includes providing (**101**) a waveguide having first and second substrates sandwiching a layer of optical recording material. As described above, such waveguides can be manufactured using a variety of different methods. In some embodiments, the layer of optical recording material contains a recorded grating. In other embodiments, the optical recording material is uncured. A forming material can be applied (**102**) to one or both surfaces of the waveguide. Various types of forming materials and application processes can be utilized. For example, inkjetting, spin coating, dip coating, or any other coating process can be utilized. As can readily be appreciated, the application technique can largely depend on the type of material utilized. Different thicknesses of forming material can be applied as appropriate depending on the application. Depending on the type of material, thicker layers can result in more haze for waveguide operation. In several embodiments, the layer of forming material applied is less than $\sim 10\ \mu\text{m}$. In further embodiments, the layer of forming material applied is less than $\sim 5\ \mu\text{m}$. A forming element having a forming surface can be provided (**103**). Various types of forming elements can be utilized. In many embodiments, the forming element can be an optical flat or a more general refractive optical element. The forming element can be brought into physical contact (**104**) with the forming material. In some embodiment, a process is applied to settle the forming element into a desired position. For example, a vacuum process can be applied to remove air pockets. The forming material can be cured (**105**) while it is in contact with the forming element. In various embodiments, curing of the planarization material (forming material) can include the application of radiation of one or more wavelengths to the material. In many embodiments, radiation of two or more wavelengths is utilized. In several embodiments, ultraviolet radiation can be used. In some embodiments, infrared radiation or visible band light can be used. Sequential or simultaneous curing steps based on any of the above procedures may be used. Curing can include thermal processes which may include infrared heating. In a number of embodiments, the curing process can include the application of pressure. In some embodiments, curing of the forming material can be carried out in a vacuum, a gas, or in a liquid. The curing process can include exposure to a humid atmosphere. The planarized waveguide can be released (**106**) from the forming element. In many embodiments, the forming material now includes a surface profile corresponding to the surface of the forming element. In various embodiments, the planarization process does not include the use of a forming element. In such cases, the forming material is cured after it is applied to the substrate or waveguide surface. The forming material can be chosen such that gravity is sufficient for planarization. In a number of embodiments, the forming element contains a previously applied release layer. The coating can be applied using spray coating of

polytetrafluoroethylene (PTFE) or similar compounds. In several embodiments, the release coating can be a fluorocarbon silylating agent. To avoid wear of the forming element, a hard coating can be applied to the forming surface of the forming element prior to application of a release coating. In some embodiments, a release coating may be applied to the forming material. Release layers and hard coating layers can be applied using various techniques, including but not limited to printing and coating techniques.

(22) After the curing process, the external surfaces of the waveguides can be coated with a hard coat. In a number of embodiments, the hard coating process replaces the planarization step with the hard coat acting as the forming material. In some embodiments, the hard coat is a liquid resin material or an acrylate. In many embodiments, the hard coat implemented is similar to the ones used to protect plastic lenses. One example of a hard coating material is the Nidek Co., Ltd (Japan) “Acier” hard coating material, which is designed for hard coating lenses and transparent plastic substrates. The coating thickness can vary widely. In some embodiments, the coating thickness is about 30 micrometers. As can readily be appreciated, the precise coating thickness can depend on the average surface non-uniformity and shape. In several embodiments, exposure of the waveguide gratings is carried out after all of the above steps have been completed. In many embodiments, exposure of the waveguide gratings can be carried out before the application of the forming material (or hard coat material). In such cases, it can be desirable that the forming material and the application process are compatible with the waveguide materials. For example, the forming material can be chosen such that its application process occurs at a temperature low enough to prevent any deformation of any formed waveguide gratings. As can readily be appreciated, the choice of forming material can depend on the material within which the waveguide gratings are formed.

(23) In many embodiments, planarization involves the application of a reference flat (made from various materials). In some embodiments, the planarization process can employ apparatus and process steps for controlling the wedge of the planarized surface. In several embodiments, surface flatness can be monitored during the planarization process. In various embodiments, the planarization process can employ an apparatus for monitoring at least one selected from the group of thickness, composition, and uniformity of the forming material during deposition. In a number of embodiments, the forming element can be cut to match the dimensions of the waveguide. In many embodiments, a tool fixture for presenting the working surface of the forming element to the forming material layer can be provided.

(24) FIGS. 2A-2E conceptually illustrate a series of process steps for planarizing a waveguide in accordance with an embodiment of the invention. FIG. 2A shows a cross section of a first waveguide substrate **200** in the XY plane **201**. In the illustrative embodiment, the surface of the substrate **200** has a deviation ΔY in the XY plane **201** from its nominal planar geometry as represented by the plot **202**. In typical cases, the deviation ΔY will vary across the surface of the substrate **200**. FIG. 2B shows a coating **203** of holographic recording material applied to the first substrate **200**. FIG. 2C shows a second substrate **204** placed over the holographic recording material layer **203**. FIG. 2D shows a layer of forming material **205** applied over the second substrate **204**. As described above, either or both outer surfaces of the waveguide can be sequentially or simultaneously planarized. FIG. 2E shows a forming element **206** with a forming surface **207** applied to the forming material **205**. In many embodiments, the forming element has a degree of planarization less than 98% wherein the degree of planarization is defined as $[1 - (T'/T)] \times 100\%$ where T is the surface deviation of the untreated surface and T' is the resulting surface deviation after completion of the planarization process.

(25) In several embodiments, the forming element is supported by a transparent substrate providing an optical path from one or more sources of curing radiation. In various embodiments, the forming element is a transparent refracting element with at least one surface having optical power for controlling the intensity distribution of the radiation presented to the forming material layer. In

many embodiments, the forming element can be an optical flat or a more general refractive optical element. In such embodiments, the curing radiation can be introduced to the forming material via the forming element. In some embodiments, the curing radiation is provided by a group of one or more UV sources distributed above a transparent forming element. FIG. 3 conceptually illustrates a curing configuration utilizing a curing source disposed above a transparent forming element in accordance with an embodiment of the invention. As shown, a waveguide **300** supports a forming material coating **301** that is in contact with a forming element **302**. In the illustrative embodiment, a curing source **303** provides curing radiation **304** that is transmitted in a diverging beam via the forming element **302** to irradiate the forming material coating **301**. In some embodiments, the curing radiation can be coupled from one or more sources spatially distributed around the forming element into an internal reflection path within the forming element. FIG. 4 conceptually illustrates a curing configuration utilizing a curing source optically coupled to a forming element in accordance with an embodiment of the invention. As shown, a waveguide **400** supports a forming material coating **401** that is in contact with a forming element **402**. Curing radiation from a curing source **404** is optically coupled to the waveguide **400** and is introduced to the forming material coating **401** via total internal reflection paths **405**. In several embodiments, the curing radiation is provided by sources spatially distributed around the waveguide. FIG. 5 conceptually illustrates a curing configuration utilizing a curing source optically coupled to a waveguide in accordance with an embodiment of the invention. As shown, a waveguide **500** supports a forming material coating **501** that is in contact with a forming element **502**. Curing radiation from a curing source **504** is optically coupled to the waveguide **500** and is introduced to the forming material coating **501** via total internal reflection paths **505**.

(26) Although FIGS. 1-5 illustrate specific processes for planarizing the surface of substrates and waveguides, such processes can be modified as appropriate depending on the specific requirements of a given application. Modifications of planarization processes in accordance with various embodiments of the invention can include the deletion or addition of various steps. For example, in some embodiments, a forming element is not needed as the forming material is capable of forming a desired flatness profile passively using gravitational forces.

(27) Planarization Materials

(28) Many different materials can be used to provide a forming or planarization material. For example, photoresists, resins, polymers, thermosets, thermoplastic polymers, polyepoxies, and polyamides may be used. In a number of embodiments, ethylene glycol diacrylate may be used. In many embodiments, the forming material has low shrinkage. Where use of a low shrinkage material is not practical, shrinkage of the planarization material can be compensated by modifying the topography of the forming surface. In some embodiments, low viscosity forming materials may be used. In several embodiments, low viscosity materials with viscosity below 1 cps (centipoise) at 20° C. may be used. In many embodiments, low viscosity materials with viscosity below 2 cps at 20° C. As can readily be appreciated, various materials having a wide range of viscosity can be utilized as appropriate depending on the specific requirements of a given application. For example, in some embodiments, the forming material is applied using deposition/printing techniques such as but not limited to inkjet printing. In such cases, it can be desirable for the material to have a viscosity compatible with the techniques used. In several embodiments, the material has a viscosity between ~10~15 cps at 20° C.

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(29) While the above description contains many specific embodiments of the invention, these should not be construed as limitations on the scope of the invention, but rather as an example of one embodiment thereof. It is therefore to be understood that the present invention may be practiced in ways other than specifically described, without departing from the scope and spirit of the present invention. Thus, embodiments of the present invention should be considered in all

respects as illustrative and not restrictive. Accordingly, the scope of the invention should be determined not by the embodiments illustrated, but by the appended claims and their equivalents.

Claims

1. A method for manufacturing a waveguide supporting a surface relief grating, the method comprising: depositing a layer of a mixture comprising a photopolymerizable monomer and a liquid crystal on a substrate; exposing the mixture to a holographic recording beam to form a volume phase grating of alternating liquid crystal-rich and liquid crystal-depleted regions; and removing the liquid crystal from the volume phase grating to form the surface relief grating of holographic photopolymer.
2. The method of claim 1, further comprising applying a surface forming process to a surface of the surface relief grating.
3. The method of claim 2, wherein applying the surface forming process comprises: applying a forming material coating to the surface relief grating; providing a forming element having a forming surface; physically contacting the forming element with the forming material coating; curing the forming material coating while it is in contact with the forming element; and releasing the forming material coating from the forming element.
4. The method of claim 3, wherein the surface forming process provides surface planarization of the surface relief grating.
5. The method of claim 4, wherein applying the surface forming process further comprises depositing at least one of a release layer or a hard coat layer.
6. The method of claim 5, wherein the release layer is a fluorocarbon silylating agent.
7. The method of claim 4, wherein curing the forming material coating comprises applying UV curing radiation via a curing configuration selected from the group consisting of: one or more UV sources distributed above the forming material coating, a fixture comprising UV sources spatially distributed around the waveguide, and one or more UV sources coupled into an internal reflection path within at least one of the forming element or the surface relief grating.
8. The method of claim 4, wherein curing the forming material coating comprises a curing configuration selected from the group consisting of: applying radiation having more than one wavelength, applying UV radiation having more than one wavelength within the UV spectrum, applying a thermal process, applying pressure, immersing said forming material coating in a vacuum, immersing said forming material coating in a gas, immersing said forming material coating in a liquid, applying radiation having a wavelength within the visible spectrum, applying radiation having a wavelength within the infrared spectrum, and exposing the forming material coating to a humid atmosphere.
9. The method of claim 4, wherein the forming material coating comprises a material selected from the group consisting of: photoresists, resins, polymers, thermosets, thermoplastic polymers, polyepoxies, polyamides, low viscosity planarization materials, materials with viscosity between 10 and 15 cps at 20° C., materials with viscosity below 2 cps at 20° C., and ethylene glycol diacrylate.
10. The method of claim 4, wherein a forming surface of the forming element has a surface characteristic for compensating for shrinkage of the forming material coating.
11. The method of claim 4, where the surface relief grating is curved; and the forming element forming surface has a curvature matching the curvature of the surface relief grating.
12. The method of claim 4, wherein the forming surface of the forming element has a degree of planarization less than 98%.
13. The method of claim 4, wherein the forming element is an optical flat, a refractive optical element, or an optical element with at least one optical surface providing optical power.
14. The method of claim 4, wherein said forming element is supported by a substrate providing an

optical path from a light source for curing.

15. The method of claim 4, wherein the forming material coating has a refractive index higher or lower than the holographic photopolymer.

16. The method of claim 1, wherein the substrate is a plastic.

17. The method of claim 1, wherein the substrate has a birefringence specified to provide smoothing of non-uniformity after a predefined number of total internal reflection (TIR) beam reflections.

18. The method of claim 1, wherein depositing the layer of the mixture onto the substrate comprises: depositing said layer of the mixture onto the substrate using at least one deposition head, wherein the mixture deposited over a grating region is formulated to achieve a predefined grating characteristic selected from the group consisting of: refractive index modulation, refractive index, birefringence, liquid crystal director alignment, grating layer thickness, and a spatial variation of the characteristic.

19. The method of claim 18, further comprising depositing at least one of a release layer or a hard coat layer using the at least one deposition head.

20. The method of claim 1, wherein the layer of the mixture comprises a polymer dispersed liquid crystal mixture.
